3 Fitting an E/I LDS model

Following the previous section, we can constraint A to obey Dale's law and C to be non-negative block diagonal to capture E/I dynamics. It is possible to implement this using constrained optimizations in the M steps when optimizing A and C. This seems to work in practice, I initialize them by learning a J from the data and then performing NNMF on it as discussed in the previous section.

TODO: Add constrained fitting results here

3.1 Inferring model parameters of a constrained E/I LDS model

We will use EM to infer the parameters of our LDS model. Specifically, we want to learn the emission parameters including the emission matrix $C \in \mathbb{R}^{N\times D}$ and the emission noise covariance $R \in \mathbb{R}^{N\times N}$. We also want to learn the dynamics parameters: the dynamics matrix $A \in \mathbb{R}^{D\times D}$, and the noise covariance matrix Q.

E step: The E step for our cell-type specific LDS remains the same as a standard LDS which uses kalman filtering and smoothening to learn a posterior over the latents $p(\mathbf{x}_t \mid y_{1:T})$. These distributions are Gaussians thankfully. Let's define $\mathbf{m}_t$ as the mean of this Gaussian distribution (for the latent $\mathbf{x}_t$) to be $\mathbf{m}_t$, and it's covariance to be Σ_t. Let's also define the cross covariance between $\mathbf{x}_t$ and $\mathbf{x}_{t+1}$ as $\Sigma_{t,t+1}$.

M step: Now, in this cell-type specific model, we require certain conditions to be true of the emission and dynamics matrices as shown in Fig. 7. To ensure that our inference procedure allows for these, we will modify the M-step.

The M step involves updating the model parameters A, C, b, d, Q, R by maximizing the expected log-likelihood of the data under our model $\mathcal{L}$, the expectation is over $P(\mathbf{x}_t \mid y_{1:T})\ \forall\ t$. For a typical LDS model, the M step is exact where A is computed as follows:

$$\left(\sum_{t=1}^{T} \mathbf{m}_t\mathbf{m}_{t+1}^\top + \Sigma_{t,t+1}\right)^\top = A\sum_{t=1}^{T-1}\left(\mathbf{m}_t\mathbf{m}_t^\top + \Sigma_t\right) \tag{71}$$

$$M_\Delta^\top = AM_{1,T-1} \tag{72}$$

where $\mathbf{m}_t$ is the mean of $P(x_t \mid y_{1:T})$ and Σ_t is the covariance of $P(x_t \mid y_{1:T})$. Similarly C is computed as follows:

$$\left(\sum_{t=1}^{T} \mathbf{m}_t\mathbf{y}_t^\top\right)^\top = C\sum_{t=1}^{T}\left(\mathbf{m}_t\mathbf{m}_t^\top + \Sigma_t\right) \tag{73}$$

$$\tilde{Y}^\top = CM_{1,T} \tag{74}$$

However, now that we have constraints on both A and C, we do not have closed-form updates anymore. Let's write the complete data log-likelihood for an LDS (for simplicity assuming b and d are set to 0):

$$\mathcal{L}_{CD}(\Theta) = -\frac{1}{2}\mathbb{E}[\sum_{t=1}^{T-1}\log\mathcal{N}(\mathbf{x}_{t+1}; A\mathbf{x}_t, Q) + \sum_{t=1}^{T}\log\mathcal{N}(\mathbf{y}_t; C\mathbf{x}_t, R)] \tag{75}$$

We will use the defined $M_{1,T-1}$, $M_\Delta, \tilde{Y}$ in the remaining section.

Updating dynamics parameters: Now, to update A, we would perform the following optimization:

$$\max_A -\frac{1}{2}\mathbb{E}[\sum_{t=1}^{T-1}(\mathbf{x}_{t+1} - A\mathbf{x}_t)^\top Q^{-1}(\mathbf{x}_{t+1} - A\mathbf{x}_t)] \text{ subject to sign constraints on } A \tag{76}$$

The objective can further be reduced to:

$$\max_A -\frac{1}{2}\text{Tr}[Q^{-1}AM_{1,T-1}A^\top - 2Q^{-1}AM_\Delta] \text{ subject to sign constraints on } A \tag{77}$$

For a quadratic solver such as CVXPY, we need to perform a few additional steps. First, for stability, let's define:

$$\tilde{Q} = \frac{Q^{-1}}{\max_{ij}|Q_{ij}|} \tag{78}$$

Note that substituting Q^{-1} with $\tilde{Q}$ doesn't affect our objective. Next, we perform a Cholesky decomposition of $\tilde{Q} = LL^\top$. This allows us to rewrite our objective as:

$$\max_A -\frac{1}{2}\text{Tr}[A^\top LL^\top AM_{1,T-1} - 2\tilde{Q}AM_\Delta] \tag{79}$$

subject to sign constraints on A We can now rewrite eq. 77 as a standard quadratic program:

$$\min_A \frac{1}{2}\text{vec}(A)^\top (I \otimes L)(M_{1,T-1} \otimes I)\left(I \otimes L^\top\right)\text{vec}(A) - \text{vec}\left(\tilde{Q}^\top M_\Delta^\top\right)^\top \text{vec}(A) \tag{80}$$

$$\text{subject to sign constraints on } A \tag{81}$$

This uses the following two vector / matrix identities:

$$\text{Tr}(XY) = \text{vec}(X^\top)^\top\text{vec}(Y) \tag{82}$$

$$\text{vec}(XY) = (Y^\top \otimes I)\text{vec}(X) \tag{83}$$

We can solve this quadratic program for a fixed Q to obtain A. And then update Q per the usual LDS update rule (as Q does not have any constraints). We set sign constraints on A such that the diagonal is unconstrained, columns corresponding to E latents are ≥ 0, and those corresponding to I latents are ≤ 0.

In presence of external inputs: Now the latent dynamics evolves as follows, while we assume the emission-latent mapping does not change.

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + B\mathbf{u}_t + \epsilon_t;\ \epsilon_t \sim \mathcal{N}(0, Q) \tag{84}$$

$$\mathbf{y}_t = C\mathbf{x}_t + \delta_t;\ \delta_t \sim \mathcal{N}(0, R) \tag{85}$$

As in case of a standard LDS, we can optimize A and B jointly. Let $\tilde{A} = \left[A, B\right]$

$$\min_{\tilde{A}} \frac{1}{2}\text{vec}\left(\tilde{A}\right)^\top (I \otimes L)\left(\tilde{M}_{1,T-1} \otimes I\right)\left(I \otimes L^\top\right)\text{vec}\left(\tilde{A}\right) - \text{vec}\left(\tilde{Q}^\top \tilde{M}_\Delta^\top\right)^\top \text{vec}\left(\tilde{A}\right) \tag{86}$$

$$\text{subject to sign constraints on } A \tag{87}$$

where $\tilde{M}_{1,T-1} = \begin{bmatrix} M_{1,T-1} & \sum_{t=1}^{T-1}\mathbf{m}_t\mathbf{u}_t^\top \\ \sum_{t=1}^{T-1}\mathbf{u}_t\mathbf{m}_t^\top & \sum_{t=1}^{T-1}\mathbf{u}_t\mathbf{u}_t^\top \end{bmatrix}$ and $\tilde{M}_\Delta = \begin{bmatrix} M_\Delta \\ \sum_{t=1}^{T-1}\mathbf{u}_t\mathbf{m}_{t+1}^\top \end{bmatrix}$.

Updating emission parameters: Next, to update C, we perform a similar optimization procedure:

$$\max_C -\frac{1}{2}\mathbb{E}[\sum_{t=1}^{T}(\mathbf{y}_t - C\mathbf{x}_t)^\top R^{-1}(\mathbf{y}_t - C\mathbf{x}_t)] \text{ s.t } C \text{ is block diagonal with blocks } C_e \text{ and } C_i;\ C_e >= 0,\ C_i >= 0 \quad (88)$$

Here, $C_e \in \mathbb{R}^{N_e \times d_e}$ will correspond to rows for E cells, and $C_i \in \mathbb{R}^{N_i \times d_i}$ will correspond to rows for I cells. The above equations can further be reduced to:

$$\max_C -\frac{1}{2}\text{Tr}[R^{-1}CM_{1,T}C^\top - 2R^{-1}C\tilde{Y}] \text{ subject to block structure on C} \quad (89)$$

Assuming R as diagonal, this reduces to:

$$\max_C -\frac{1}{2}\text{Tr}\left(\sum_{j=1}^{N}\left(r_j^{-1}c_j^\top c_j\right)M_{1,T}\right) + \sum_{j=1}^{N} r_j^{-1}c_i\tilde{y}_j \quad (90)$$

where $c_j \in \mathbb{R}^{1\times d}$ is the $j-$th row of C, and $\tilde{y}_i \in \mathbb{R}^{d\times 1}$ is the $j-$th column of $\tilde{Y}$. This expression will allow us to now learn the corresponding row of C for each neuron separately. Writing this as a standard quadratic program (for CVXPY), for each row of C, we get:

$$\min_{\mathbf{c}_j} \frac{1}{2}\mathbf{c}_j M_{1,T}\mathbf{c}_j^\top - \mathbf{c}_j\tilde{\mathbf{y}}_j \quad (91)$$

where for E cells $\mathbf{c}_j[: d_e] \geq 0, \mathbf{c}_j[d_e :] = 0$, and for I cells $\mathbf{c}_j[d_e :] \geq 0, \mathbf{c}_j[: d_e] = 0$ R can be updated per the usual LDS update rule.

3.1.1 Learn signs within the M step

In case that the identity of a neuron is unknown, we would like to be able to infer it as well while also learning the model parameters. Our goal in this setting would be to learn both the emission matrix C, as well as the identities of the unknown neurons.

For each unknown neuron, we can optimize eq. 91 twice assuming the neuron to be excitatory and inhibitory, and then choose the identity that results in the minimum objective.

3.2 Non-identifiability in the model

$$x_{t+1} = Ax_t + \epsilon_t \quad (92)$$

$$y_t = Cx_t + \delta_t \quad (93)$$

Let $S \in \mathbb{R}^{D\times D}$ be a diagonal matrix with positive entries. Then, we can find an equivalent set of parameters describing the same data such that:

$$x_{t+1} = \hat{A}x_t + \hat{\epsilon}_t \quad (94)$$

$$y_t = \hat{C}x_t + \hat{\delta}_t \quad (95)$$

$$\hat{C} = CS,\ \hat{A} = S^{-1}AS,\ \hat{Q} = S^{-1}QS^{-1},\ \hat{R} = R,\ \hat{\Sigma}_0 = S^{-1}\Sigma_0 S^{-1} \quad (96)$$

where Σ_0 is the covariance of the initial state distribution.